



INSTITUTE OF AERONAUTICAL ENGINEERING (Autonomous)

Dundigal, Hyderabad - 500 043

QUESTION BANK

Course Title	Environmental and Sustainable Development				
Course Code	AMCE01				
Program	B.Tech				
Semester	III	CSE			
Course Type	Mandatory				
Regulation	IARE - BT25				
Course Structure	Theory			Practical	
	Lecture	Tutorials	Credits	Laboratory	Credits
	1	-	1	-	-
Course Coordinator	Dr.M.S.N.A.prasad , Associate Professor				

COURSE OBJECTIVES:

The students will try to learn:

I	The interrelationship between living organism and environment.
II	The importance of environment by assessing its impact on the human world
III	The knowledge on themes of biodiversity, natural resources, pollution control and waste management.
IV	The sustainability and unsustainability of various interactions between human society and the earth's natural systems'

COURSE OUTCOMES:

After successful completion of the course, students should be able to:

CO 1	Explain the basic ecological principles, biogeochemical cycles and its function for the flow of energy in ecosystem
CO 2	Awareness on different natural resources and its conservation for sustainable development
CO 3	Use alternate energy resources for future growing energy needs.
CO 4	Predict the importance of biodiversity for its productive use.

CO 5	Identify the global environmental problems by different types of environmental Pollution and international summits for minimizing the problems
CO 6	Outline the features of laws and rules related to environment protection, environmental impact assessment towards sustainable development

MODULE I				
ECOSYSTEMS				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1	Explain energy flow pattern in different types of ecosystem. What happens to the energy flow as we move up the tropical levels	Understand	Learner to recall energy flows in a unidirectional manner through all ecosystems, beginning with the Sun as the primary source of energy. Producers capture solar energy through photosynthesis and convert it into chemical energy.	CO1
2	Explain the major drawbacks of using pesticides and other agrochemicals in the soil on any ecosystem.	Understand	Learner to recall the excessive use of pesticides and other agrochemicals has several harmful effects on the soil and the environment.	CO1
3	Why ecosystem is getting degraded and what are the steps have to be taken by public?	Understand	Learner to recall Learner to recall types of of ecosystems and energy flow in ecosystems	CO1
4	If every organism must eat another organism for survival, where does the food chain start	Understand	learner to recall that every organism depends on another organism for food and survival, forming a food chain. A food chain always starts with producers which prepare their own food through photosynthesis using sunlight water and carbon dioxide.	CO1

5	Mention type of cycles we have in an ecosystem. Explain how water circulates on earth.	Understand	Learner to recall ecosystems contain different biogeochemical cycles that recycle essential nutrients between living organisms and the environment. The major cycles include the water cycle carbon cycle nitrogen cycle oxygen cycle and phosphorus cycle.	CO1
6	What is the scope and importance of environment	Understand	The learner recalls that the environment includes all the living and non-living components that surround us and interact with one another.	CO1
7	Explain why there are only four to five trophic levels in any ecosystem.	Understand	Learner to recall that an ecosystem usually has only four to five trophic levels because only a small portion of energy is transferred from one trophic level to the next, while most of the energy is lost as heat and through metabolic activities.	CO1
8	Write a short note on the need for public awareness on environment in today's scenario	Understand	learner recalls that public awareness about the environment is essential to address current environmental challenges such as pollution, climate change, deforestation, biodiversity loss, and waste generation.	CO1

9	Explain how food becomes energy in organisms.	Understand	Learner to recall that food becomes energy through the process of cellular respiration. During digestion, complex food substances are broken down into simple molecules such as glucose. In the cells, glucose combines with oxygen to release energy, which is stored in the form of ATP	CO1
10	What is the role of an individual to conserve the environment?	Understand	Learner to recall every individual plays an important role in conserving the environment through responsible actions and sustainable practices.	CO1
PART B-LONG ANSWER QUESTIONS				
1	Explain energy flow pattern in different types of ecosystem	Understand	Learner to recall energy flows in a unidirectional manner through all ecosystems, beginning with the Sun as the primary source of energy. Producers capture solar energy through photosynthesis and convert it into chemical energy.	CO1
2	List the main components of ecosystem and briefly describe the functions of each	Understand	Learner to recall an ecosystem consists of two main components biotic and abiotic components. biotic components include producers, consumers, and decomposers. Producers prepare food through photosynthesis	CO1

3	Explain about carbon and nitrogen cycles with the help of a diagram.	Understand	Learner to recall the carbon cycle describes the movement of carbon between the atmosphere, living organisms soil oceans and fossil fuels. Plants absorb carbon dioxide from the atmosphere during photosynthesis	CO1
4	Discuss the significance of food chains and food webs with relevant examples.	Understand	Learner to recall that food chains and food webs show the transfer of energy and nutrients among organisms in an ecosystem. A food chain represents a simple pathway of energy flow from producers to consumers and decomposers while a food web shows the interconnected food relationships among different organisms.	CO1
5	What are biogeochemical cycles? Explain phosphorous cycle with the help of a diagram.	Understand	Learner to recall the biogeochemical cycles are natural processes through which essential elements and compounds move between living organisms and the non-living environment . These cycles help recycle nutrients and maintain the balance of ecosystems.	CO1
6	Name all the cycles that constitute the proper functioning of an ecosystem	Understand	Learner to recall the biogeochemical cycles are natural processes through which essential elements and compounds move between living organisms and the non-living environment . These cycles help recycle nutrients and maintain the balance of ecosystems	CO1

7	Explain about the biomagnification property of a food chain	Understand	Learner to recall that biomagnification is the process in which the concentration of harmful non-degradable substances such as pesticides and heavy metals increases as they move through successive levels of a food chain.	CO1
8	What are ecological pyramids? Explain why some of these pyramids are upright while others are inverted in different ecosystem.	Understand	Learner to recall that ecological pyramids are graphical representations of the relationship between organisms at different trophic levels in an ecosystem. They show the distribution of number of organisms, biomass, or energy at each level of a food chain.	CO1
9	Differentiate between grazing food chain and detritus food chain.	Understand	Learner to recall grazing food chain and detritus food chain are two major types of food chains found in ecosystems. A grazing food chain begins with living green plants which are eaten by herbivores and then by carnivores. It represents the direct transfer of energy from plants to consumers.	CO1
10	List the different trophic levels of ecosystem?	Understand	The learner recalls that trophic levels represent the different feeding positions of organisms in an ecosystem based on the transfer of energy through a food chain. The main trophic levels include primary consumers secondary consumers tertiary consumers and decomposers	CO1

11	Explain the role of producers, consumers and decomposers in an ecosystem with practical example.	Understand	The learner recalls that trophic levels represent the different feeding positions of organisms in an ecosystem based on the transfer of energy through a food chain. The main trophic levels include primary consumers secondary consumers tertiary consumers and decomposers	CO1
12	What services do ecosystems provide us	Understand	Learner to recall ecosystems provide essential services that support life and human well-being ecosystem services include provisioning services such as food water timber and medicinal resources	CO1
13	Differentiate among species, population and community.	Understand	Learner to recall understands that species form populations, and different populations together form communities, which interact with the physical environment to make up an ecosystem.	CO1
14	List the different tropic levels of ecosystem?	Understand	The learner recalls that trophic levels represent the different feeding positions of organisms in an ecosystem based on the transfer of energy through a food chain. The main trophic levels include primary consumers secondary consumers tertiary consumers and decomposers	CO1

15	What is the role of an individual to conserve the environment	Understand	Learner to recall individuals play an important role in conserving the environment through responsible actions and sustainable practices. These include saving water and energy, reducing waste, reusing and recycling materials, planting and protecting trees avoiding pollution using eco-friendly products and promoting environmental awareness.	CO1
16	Write a short note on 1. Single channel energy model 2. Y shaped energy model	Understand	Learner to recall models help explain the movement of energy feeding relationships and the role of different organisms in maintaining ecosystem balance.	CO1
17	Explain about hydrological cycle with the help of a diagram.	Understand	Learner to recall the hydrological cycle is the continuous movement of water between the Earth surface atmosphere, and underground regions. Water from oceans, rivers, lakes, and plants enters the atmosphere through evaporation and transpiration.	CO1
18	Define Eco system? Explain the scope and importance of ecosystem	Understand	Learner to recall that ecosystems are important because they provide essential services such as food water oxygen production climate regulation soil formation, and nutrient recycling	CO1

19	Discuss the need for public awareness on environment.	Understand	Learner to recall public awareness about the environment is essential for understanding and addressing environmental issues such as pollution climate change deforestation loss of biodiversity and improper waste disposal	CO1
20	Explain the major drawbacks of using pesticides and other chemicals in the water ecosystem.	Understand	Learner to recall use of pesticides and other chemicals can negatively affect water quality and aquatic ecosystems. These chemicals can enter rivers lakes and groundwater through runoff and leaching, causing water pollution	CO1
PART C-SHORT ANSWER QUESTIONS				
1	Define ecosystem. How does an ecosystem work.	Understand	The learner recalls that an ecosystem works through the interaction of living and non-living components. Producers capture solar energy and prepare food, consumers obtain energy by feeding on other organisms	CO1
2	What are biotic and abiotic factors	Understand	The learner recalls that an ecosystem works through the interaction of living and non-living components. Producers capture solar energy and prepare food, consumers obtain energy by feeding on other organisms	CO1

3	Explain briefly the importance of ecological pyramids.	Understand	learner recalls ecological pyramids are important tools used to represent the relationship between different trophic levels in an ecosystem. They show the number of organisms biomass and energy distribution at each feeding level	CO1
4	What are producers consumers and decomposers in an ecosystem	Understand	Learner to recall the interaction among producers consumers and decomposers helps maintain energy flow nutrient recycling and ecological balance in an ecosystem.	CO1
5	Discuss pyramid of biomass	Understand	Learner to recall The learner understands that the pyramid of biomass helps explain the flow of energy, feeding relationships, and distribution of living matter among organisms in an ecosystem. In some aquatic ecosystems, the pyramid of biomass may be inverted because phytoplankton have a lower standing biomass but reproduce rapidly and support larger consumer populations.	CO1
6	What is meant by biomagnification	Understand	The learner recalls that biomagnification is the process in which the concentration of harmful, non-biodegradable substances such as pesticides and heavy metals increases at each successive trophic level of a food chain.	CO1

7	Explain the energy flow of an ecosystem.	Understand	Learner to recall energy flow in an ecosystem is a one-way process that begins with the Sun. Producers such as green plants capture solar energy through photosynthesis and convert it into chemical energy.	CO1
8	Discuss pyramid of energy.	Understand	Learner to recall the pyramid of energy is a graphical representation of the amount of energy available at each trophic level in an ecosystem.	CO1
9	What is the nature of an ecosystem Give its structure and functions.	Understand	Learner to recall ecosystems may be natural or artificial and they vary in size from small ecosystems like ponds to large ecosystems like forests and oceans. The interactions among organisms and their environment help maintain ecological balance and support life.	CO1
10	Differentiate between food chain and food web.	Understand	The learner to recall understands that a food chain shows a single route of energy transfer while a food web shows multiple interconnected routes making ecosystems more stable and balanced.	CO1
11	Discuss pyramid of number.	Understand	Learner to recall that pyramid of numbers helps explain population distribution and feeding relationships among organisms in an ecosystem.	CO1

12	What is the structure of an ecosystem?	Understand	Learner to recall the interaction between biotic and abiotic components forms the functional structure of an ecosystem and helps maintain ecological balance.	CO1
13	Relate ecosystem and biosphere	Understand	Learner to recall that ecosystems are parts of the biosphere, and the combined functioning of all ecosystems makes up the biosphere.	CO1
14	What are the processes that autotrophic organisms use to produce organic material from inorganic substances	Understand	learner to recall autotrophs are primary producers in ecosystems and form the base of food chains by converting inorganic substances into organic matter, which provides energy for other organisms.	CO1
15	Classify ecosystems based on their habitat, creation and exchange of matter and energy.	Understand	learner to recall ecosystems differ in structure and function but all depend on interactions between organisms and their physical environment for maintaining ecological balance.	CO1
16	List the different trophic levels of ecosystem.	Understand	Learner to recall energy decreases as it moves from one trophic level to the next, and each level plays an important role in maintaining ecosystem balance	CO1
17	What kind of organisms do we find in aquatic life zones?	Understand	Learner to recall organisms interact with each other and their aquatic environment, maintaining energy flow, nutrient cycling, and ecological balance.	CO1

18	Define biogeochemical cycles. Explain their importance.	Understand	Learner to recall the biogeochemical cycles help in nutrient recycling maintaining ecosystem balance and supporting life processes on Earth	CO1
19	What are terrestrial and aquatic ecosystems	Understand	learner to recall understands that both terrestrial and aquatic ecosystems support biodiversity, maintain energy flow, recycle nutrients, and contribute to the balance of the biosphere.	CO1
20	Define autotrophic and heterotrophic organisms.	Understand	Learner to recall that autotrophs provide energy for the ecosystem, while heterotrophs transfer and recycle energy and nutrients through food chains and food webs.	CO1

MODULE II

NATURAL RESOURCES

PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS

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1	Discuss the importance of land as a natural resource. Explain how world's land surface is getting degraded.	Understand	Learner to recall land helps in soil formation, nutrient cycling, water regulation, and maintaining ecosystem balance.	CO2
2	Explain how every source of energy has its limits. What should be our attitude to using energy in this world	Understand	Learner to recall land helps in soil formation, nutrient cycling, water regulation, and maintaining ecosystem balance.	CO2
3	What changes would you incorporate in your life-style in a move towards a sustainable use of our resources?	Understand	Learner to recall land helps in soil formation, nutrient cycling, water regulation, and maintaining ecosystem balance.	CO2

4	What you can do to conserve water? Explain how rain water can be harvested	Understand	Learner to recall water conservation involves the careful use protection and management of water resources to prevent wastage and ensure availability for future generations.	CO2
5	Water is a unique resource. Comment on this statement	Understand	Learner to recall water is a unique natural resource because it is essential for the survival of all living organisms and plays a vital role in maintaining ecosystems.	CO2
6	What are the environmental and social impacts of mining?	Understand	Learner to recall mining activities involve the extraction of minerals and resources from the Earth, which can have significant environmental and social impacts.	CO2
7	What are the major anthropogenic causes of droughts and floods? Give its remedial measures.	Understand	Learner to recall anthropogenic causes are human activities that contribute to environmental changes, including droughts and floods.	CO2
8	What is wind energy? Enumerate advantages and disadvantages associated with wind power.	Understand	Learner to recall wind power is a renewable source of energy obtained by converting the kinetic energy of moving air into electrical energy using wind turbines.	CO2
9	Discuss some of the water resources problems in India.	Understand	Learner to recall India faces several challenges related to water resources due to increasing demand, population growth, and environmental changes.	CO2

10	What kind of energy do we get from the sun? Explain how we convert solar energy in to electricity.	Understand	Learner to recall Sun is the primary source of energy for Earth and provides solar energy in the form of light and heat energy.	CO2
PART B-LONG ANSWER QUESTIONS				
1	Classify natural resources. What are the different methods of conserving water resource?	Understand	Learner to recall different methods of conserving water resources by identifying and listing common water conservation practices such as rainwater harvesting preventing water wastage repairing leaks reusing water using efficient irrigation methods, protecting water bodies from pollution, and planting trees to help recharge groundwater	CO3
2	Explain how water becomes a renewable resource. What are the effects of over exploitation of water resources?.	Understand	Learner to recall how water becomes a renewable resource by understanding that water is continuously renewed through the water cycle.	CO3
3	Discuss briefly droughts and floods with respect to their occurrence and impacts.	Understand	Learner recall droughts and floods with respect to their occurrence and impacts by identifying the causes and conditions that lead to these natural disasters.	CO3

4	What are the advantages and disadvantages of dams to the society and environment?	Understand	Learner to recall the advantages and disadvantages of dams to society and the environment by identifying their benefits, such as irrigation hydroelectric power generation flood control and water supply as well as their drawbacks including displacement of people, loss of forests and wildlife habitats disruption of river ecosystems and environmental pollution.	CO3
5	Discuss in detail the major environmental and social impacts of mineral extraction..	Understand	Learner to recall the major environmental and social impacts of mineral extraction by identifying its effects on land water air forests and biodiversity.	CO3
6	Explain how serious are water logging and soil salinity in land degradation	Understand	Learner to recall how waterlogging and soil salinity contribute to land degradation by explaining that excessive irrigation and poor drainage cause water to accumulate in the soil and increase salt levels.	CO3
7	List out alternate energy sources. Explain their present status, merits and demerits.	Understand	Learner to recall alternate energy sources by identifying renewable and non-conventional sources of energy such as solar energy, wind energy, hydroelectric energy, geothermal energy, tidal energy, and biomass energy	CO3

8	What is a wind farm? Enumerate advantages and disadvantages associated with wind power.	Understand	Learner to recall wind farm is and enumerates the advantages and disadvantages associated with wind power by defining a wind farm as a collection of wind turbines installed in a windy location to generate electricity	CO2
9	Compare the advantages and disadvantages of oil, coal and natural gas as energy sources.	Understand	Learner to recall advantages and disadvantages of oil coal and natural gas by identifying their uses as fossil fuels for energy production transportation and industries.	CO2
10	What are the advantages of hydropower? How energy is generated in hydroelectric power plant.	Understand	Learner to recall advantages of hydropower by identifying it as a renewable source of energy generated from flowing or falling water.	CO3
11	Explain the cause of over exploitation of natural resources. Why do we need to manage our resources?	Understand	Learner to recall causes of overexploitation of natural resources by identifying factors such as rapid population growth, increasing industrialization urbanization excessive consumption agricultural expansion deforestation mining activities and improper management of resources.	CO3
12	What are the major anthropogenic causes of droughts and floods? Give its remedial measures.	Understand	Learner to recall major anthropogenic causes of droughts and floods by identifying human activities that contribute to these events.	CO3

13	Explain how solar energy converted to electrical power and what are the problems in this regard.	Understand	Learner to recall how solar energy is converted into electrical power by explaining that solar panels contain photovoltaic (PV) cells that absorb sunlight and convert solar energy directly into electrical energy.	CO3
14	What is the impact of urbanization and industrialization on land	Understand	Learner to recall impact of urbanization and industrialization on land by identifying how the expansion of cities industries and infrastructure affects land resources.	CO3
15	Write a note on non-renewable energy resources. Explain how almost every source of energy has its limits.	Understand	Learner to recall non-renewable energy resources by identifying energy sources that are limited and cannot be replenished within a short period of time.	CO3
16	Explain briefly about geo thermal energy and give its advantages and disadvantages	Understand	Learner to recall briefly about geothermal energy and gives its advantages and disadvantages by explaining that geothermal energy is the heat energy obtained from the Earth's interior and is used for electricity generation and heating purposes.	CO3
17	What are renewable and non renewable energy sources and give the advantages of renewable energy sources.	Understand	Learner to recall renewable and non-renewable energy sources by identifying the different types of energy resources and distinguishing between them.	CO3

18	What are the advantages of solar energy? Explain the role of solar energy in replacing natural energy sources.	understand	Learner to recall advantages of solar energy by identifying solar energy as a clean, renewable, and sustainable source of energy obtained from sunlight	CO3
19	What are dams? Discuss the advantages and disadvantages of dams?	understand	Learner to recall advantages and disadvantages of dams by explaining that a dam is a barrier constructed across a river or water stream to store and control water for various purposes..	CO2
20	Explain briefly about rain harvesting plant and give its importance in urban society	understand	Learner to recall briefly about rainwater harvesting plants and their importance in urban society by explaining that a rainwater harvesting plant is a system used to collect filter, and store rainwater from rooftops or other surfaces for later use or groundwater recharge.	CO3

PART C-SHORT ANSWER QUESTIONS				
1	What are renewable and non-renewable resources? Give examples..	Understand	Learner to recall renewable and non-renewable resources by identifying and differentiating between resources based on their availability and rate of replacement.	CO2
2	Enlist different surface and ground water resources.	Understand	Learner to recall surface and groundwater resources by identifying the two main types of water resources.	CO3
3	What are the environmental and social impacts of mining?	Understand	Learners identify and recall the key environmental and social impacts of mining. They recognize how mining can affect the environment through land degradation deforestation water and air pollution, and loss of biodiversity as well as its social effects including community displacement health issues employment opportunities and changes to local livelihoods.	CO3
4	What is an aquifer? Discuss its typesHow are they classified?	Understand	Learners to recal define an aquifer as an underground layer of permeable rock, sand, or gravel that stores and transmits groundwater.	CO2
5	List the environmental effects of using of mineral resources.	Understand	Learner to recall Learners identify and recall the major environmental effects associated with the use of mineral resources.	CO3
6	Define mineral. Write its uses.	Understand	Learner to recall define a mineral as a naturally occurring inorganic solid with a definite chemical composition and an ordered crystalline structure.	CO3

7	What is the importance of rain water harvesting	Understand	Learner to recall the importance of rainwater harvesting as a sustainable method of conserving water.	CO3
8	What are the benefits and problems provided by large dams?	Understand	Learners identify and recall the major benefits and problems of large dams. They recognize benefits such as irrigation, hydroelectric power generation flood control water supply and recreation.	CO3
9	Define solar cells. Write its applications	Understand	Learner to recall a solar cell as a device that converts sunlight directly into electrical energy through the photovoltaic effect.	CO3
10	What is the impact of deforestation on the environment?	Understand	Learner to recall identify the major environmental impacts of deforestation.	CO3
11	State any two reasons that why should we conserve natural resources.	Understand	Learner to recall Learners identify and recall two key reasons for conserving natural resources to ensure their sustainable availability for future generations, and to protect the environment by maintaining ecological balance conserving biodiversity and reducing pollution.	CO3
12	What is the importance of land as a natural resource	Understand	Learner to recall the importance of land as a valuable natural resource. They recognize that land provides space for agriculture forestry industries settlements, transportationand wildlife habitats.	CO3

13	List the different types of natural resources.	Understand	Learner to recall Learners identify and recall the various types of natural resources and their classifications. They recognize that natural resources are broadly classified into renewable resources and non-renewable resources	CO3
14	What are the uses of various types of minerals	Understand	Learner to recall the different types of minerals based on their properties and uses. They recognize that minerals are mainly classified into metallic minerals non-metallic minerals	CO3
15	Discuss the problems of over exploitation of ground water.	Understand	Learner to recall Learners identify and recall the major problems caused by excessive extraction of groundwater. They recognize impacts such as depletion of groundwater levels drying of wells and water sources land subsidence saltwater intrusion in coastal areas, reduced availability of water for future use, and deterioration of groundwater quality.	CO2
16	What exactly are fossil fuels and why are they non-renewable	Understand	Learner to recall fossil fuels such as coal petroleum and natural gas.	CO3

17	Define soil erosion. How can it be checked	Understand	Learner to recall the meaning causes and effects of soil erosion. They understand that soil erosion is the removal of the top fertile layer of soil by agents such as wind water and human activities.	CO3
18	What is meant by exhaustible natural resources? Mention any two alternate energy resources and give their uses.	Understand	Learner to recall exhaustible natural resources as resources that are limited in quantity and can be depleted through continuous use.	CO3
19	Describe the measures to be taken for the control of floods	Understand	Learner to recall the measures used to control and reduce the impacts of floods.	CO3
20	What are the main reasons for water scarcity in India?	Understand	Learner to recall the major causes of water scarcity in India.	CO3
MODULE III				
BIODIVERSITY AND BIOTIC RESOURCES				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1	What are the three categories in to which biodiversity can be separated	Understand	Learner to recall the three main categories of biodiversity genetic diversity species diversity and ecosystem diversity.	CO4

2	List different developmental activities, including construction of dams affect the biodiversity and action need to be taken to conserve them.	Understand	Learner to recall the impacts of developmental activities such as dam construction on biodiversity. They recognize that these activities can cause habitat destruction submergence of forests and wildlife areas, displacement of species loss of biodiversity disruption of natural ecosystems and changes in river flow patterns.	CO4
3	Why is biodiversity valuable and how can be the value of biodiversity estimated.	Understand	Learner to recall Learners identify and recall the importance and value of biodiversity. They understand that biodiversity provides ecological services economic benefits social and cultural values and resources such as food, medicines, and raw materials.	CO4
4	India is one of the world's richest countries in terms of its vast array of biological diversity. Support the statement with three facts.	Understand	Learner to recall the countries that are considered among the richest in biological diversity due to their high variety of plants animals and ecosystems. They recognize biodiversity-rich countries such as Brazil, Colombia, Indonesia, Mexico, Australia, India, and Madagascar.	CO4
5	What are the basic issues in protecting wild flora and fauna?	Understand	Learner to recall Learners identify and recall the major issues related to the conservation of wild plants and animals.	CO4

6	Describe the measures taken by India to save our biodiversity.	understand	Learner to recall major measures adopted by India for the protection and conservation of biodiversity.	CO4
7	Explain how ex-situ conservation is practised. Write its advantages and disadvantages	Understand	Learner to recall Ex-situ conservation is the practice of protecting and conserving plants and animals outside their natural habitats.	CO4
8	Describe how effective are different types of protected areas at conserving biodiversity and providing ecosystem services.	Understand	Learner to recall Protected areas are places set aside to conserve biodiversity protect natural habitats and maintain important ecosystem services.	CO4
9	Name the animals for whose protection and conservation specific projects have been launched in our country.	Understand	Learner to recall Learner to recall Name the animals for whose protection and conservation specific projects have been launched in our country.	CO4
10	Explain how in-situ conservation is practised. Write its advantages and disadvantages.	Understand	Learner to recallconserving a species in its natural habitat. Instead of picking up an animal or plant and moving it to a zoo or seed bank we protect the ecosystem they live in so they can thrive and evolve naturally.	CO4
PART B-LONG ANSWER QUESTIONS				
1	What are primary and secondary air pollutants? Enumerate various methods to control air pollution	Understand	Learner to recall the types of pollution and impacts	CO4
2	Enumerate with examples the major sources of surface water pollution and ground water pollution. Explain the methods to control water pollution.	Understand	Learner to recall the types of pollution and impacts	CO4

3	Describe the major sources of soil pollution. How does soil pollution affect soil productivity and what measures can be taken to prevent soil pollution	Understand	Learner to recall the types of pollution and impacts	CO4
4	Describe briefly the sources, effects and control of noise pollution	Understand	Learner to recall the types of pollution and impacts.	CO4
5	Enumerate any five major categories of water pollutants, their sources and effects.	Understand	Learner to recall the types of pollution and impacts	CO4
6	Discuss how e-waste can be managed to prevent environmental pollution.	Understand	Learner to recall Managing e-waste is critical because these items contain dangerous heavy metals	CO4
7	Describe the various ways of treating and purifying sewage water.	Understand	Learner to recall how to dispose waste water and consequences	CO4
8	What adverse effects can solid wastes cause? Discuss how can the solid waste be managed.	Understand	Learner to recall Solid waste includes everything we throw away in our trash bins household garbage industrial scraps construction debris and medical waste.	CO4
9	Explain the concept of bioremediation. What are the advantages of bioremediation	Understand	Learner to recall Bioremediation is an environmental cleanup technique that uses living organisms—primarily microbes (bacteria and fungi) and plants to break down, neutralize remove toxic pollutants from soil water and air.	CO4
10	Discuss Kyoto protocol. What can it do to curb climate change	Understand	Learner to recall how climate changed suddenly and their impact on humanbeings.	CO4

11	What are the major causes and consequences of deforestation?	Understand	Learner to recall Learners the major effects of deforestation on the environment wildlife and human life.	CO4
12	Discuss the difference of opinion between north block and south block countries during earth summit, 1992	Understand	Learner to recall Learners will understand and remember the different viewpoints of developed and developing countries at the Earth Summit held in Rio de Janeiro in 1992.	CO4
13	Explain climate change and global warming. How are they related and what is the evidence that proves climate is changing?.	Understand	Learner to recall Learners will understand and remember the meanings of climate change and global warming.	CO4
14	What are greenhouse gases? Discuss the potential and contribution of these gases to global warming phenomenon.	Understand	Learner to recall will understand and remember the major greenhouse gases including carbon dioxide methane nitrous oxide water vapour and fluorinated gases.	CO4
15	What are ozone depleting substances? Discuss the theme of Montreal protocol.	Understand	Learner to recall the substances that cause depletion of the ozone layer such as chlorofluorocarbons halons carbon tetrachloride methyl chloroform and methyl bromide.	CO4
16	Describe briefly waste water treatment methods?	Understand	Learner to recall the different methods used for treating wastewater before it is released into the environment or reused.	CO4

17	Discuss Montreal protocol. What can it do to curb climate change.	Understand	Learner to recall the significance of the Montreal Protocol, an international environmental agreement adopted in 1987 to protect the ozone layer.	CO4
18	Enumerate with examples the sources of surface water?	Understand	Learner to recallLearners will understand and remember the different sources of surface water available on the Earth's surface.	CO4
19	What are the major causes and consequences of desertification?	Understand	Learner to recall the main causes of desertification including deforestation overgrazing unsustainable agricultural practices climate change and improper land and water management.	CO4
20	Explain the consequences of surface and ground water?	Understand	Learner to recall Learners will understand and remember the major consequences affecting surface water and groundwater resources.	CO4
PART C-SHORT ANSWER QUESTIONS				
1	Define pollution. Name various atmospheric pollutants.	Understand	Learner to recall pollution is the introduction of harmful substances or pollutants into the environment causing undesirable changes and negatively affecting living organisms and natural resources.	CO4

2	What is noise pollution? Mention its sources	Understand	Learner to recall noise pollution refers to the presence of unwanted or excessive sound in the environment that can disturb human life and other living organisms.	CO4
3	What are the natural and man-made pollutants that cause air pollution?.	Understand	Learner to recall air pollution is the contamination of the atmosphere by harmful substances	CO4
4	Mention various methods to control air pollution in industries	Understand	Learner to recall the various methods used to control industrial air pollution.	CO4
5	Discuss soil pollution caused by fertilizers	Understand	Learner to recall that excessive and improper use of chemical fertilizers can cause soil pollution..	CO4
6	Discuss the adverse health effects due to industrial noise.	Understand	Learner to recall the harmful health effects caused by prolonged exposure to industrial noise	CO4
7	Explain how ground water get pollutes.	Understand	Learner to recall major causes of groundwater pollution.	CO4
8	Define air pollution. What are the sources of air pollution	Understand	Learner to recall Learners will understand and remember that air pollution is the contamination of the atmosphere by harmful substances such as gases smoke dust and particulate matter.	CO4

9	Define solid waste. How can solid waste be recycled	Understand	Learner to recall Learners will understand and remember that solid waste refers to unwanted or discarded solid materials generated from households, industries, agriculture, and commercial activities	CO4
10	Describe how we measure water quality	Understand	Learner to recall Learners will understand and remember the important parameters used to measure water quality.	CO4
11	Mention treatment methods used for industrial effluents	Understand	Learner to recall the different methods used for treating industrial effluents before their discharge into the environment.	CO4
12	Define e-waste. Why does it require management?	Understand	Learner to recall that electronic waste consists of discarded electrical and electronic devices such as computers, mobile phones, televisions, and household appliances.	CO4
13	What is bioremediation?	Understand	Learner to recall Learners will understand and remember that bioremediation is the process of using living organisms such as microorganisms fungi or plants to remove degrade or neutralize pollutants	CO4
14	What is ozone layer and why is it getting depleted.	Understand	Learner to recall Learners will understand and remember that the ozone layer is a region of the Earth's stratosphere that absorbs most of the Sun's harmful ultraviolet radiation.	CO4

15	What are the major causes of deforestation	Understand	Learner to recall the major causes of deforestation.	CO4
16	Outline the provisions of the Kyoto protocol.	Understand	Learner to the key provisions of the Kyoto Protocol. They will be able to identify that the protocol set legally binding greenhouse gas emission reduction targets for developed countries	CO4
17	What are ozone depleting substances?	Understand	Learner to recall the major ozone-depleting substances (ODS) responsible for the depletion of the ozone layer.	CO4
18	What are the major causes of desertification?	Understand	Learner to recall the major causes of desertification.	CO4
19	Differentiate between climate change and global warming.	Understand	Learner to recall the distinction between climate change and global warming.	CO4
20	What are the results of implementation of the Montreal protocol?	Understand	Learner to recall the major outcomes of the implementation of the Montreal Protocol.	CO4
MODULE IV				
ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1	List the physical, physiological and psychological effects of noise. Discuss the adverse health effects due to industrial noise.	Understand	Learner to recall the effects of excessive noise on human health and well-being.	CO5
2	Explain climate change and global warming. How are they related and what is the evidence that proves climate is changing?	Understand	Learner to recall the concepts of climate change and global warming.	CO5

3	What is the role of automobiles in creating air pollution and other environmental problems?	Understand	Learner to recall how automobiles contribute to air pollution and environmental degradation.	CO5
4	Enumerate any five major categories of water pollutants, their sources and effects..	Understand	Learner to recall the major categories of water pollutants and their impacts on water quality and ecosystems.	CO5
5	Explain how groundwater in many places in India become contaminates with fluoride and arsenic.	Understand	Learner to recall the causes of fluoride and arsenic contamination in groundwater.	CO5
6	Draw layouts of Sewage treatment, effluent treatment and common effluent treatment plants.	Understand	Learner to recall the basic layouts and treatment stages involved in sewage treatment plants effluent treatment plants and common effluent treatment plants	CO5
7	Explain the problems encountered in the disposal of solid waste from various sources.	Understand	Learner to recall the challenges associated with the disposal of solid waste generated from households industries agriculture healthcare facilities and commercial activities.	CO5
8	How can a change of one or two degrees in global average temperatures have an impact on our lives?	Understand	Learner to recall that even a small increase of one or two degrees in global average temperature can have significant impacts on the Earth's climate system.	CO5
9	What are the international initiatives against the depletion of the ozone layer?.	Understand	Learner to recall the major international efforts taken to protect the ozone layer.	CO5

10	Explain how does climate change become a global issue and what were the international initiatives..	Understand	Learner to recall that climate change became a global issue due to the increasing concentration of greenhouse gases from human activities such as burning fossil fuels deforestation industrialization and changes in land use.	CO5
PART B-LONG ANSWER QUESTIONS				
1	What are primary and secondary air pollutants? Enumerate various methods to control air pollution	Understand	Learner to recall Air pollution can be effectively reduced by combining cleaner technologies strict emission controls afforestation efficient transportation and strong environmental regulations along with public participation.	CO5
2	Enumerate with examples the major sources of surface water pollution and ground water pollution. Explain the methods to control water pollution	Understand	Learner to recall Surface water and groundwater are essential resources for drinking, agriculture, industry, and ecosystems.	CO5
3	Describe the major sources of soil pollution. How does soil pollution affect soil productivity and what measures can be taken to prevent soil pollution?	Understand	Learner to recall Soil pollution is mainly caused by agricultural chemicals, industrial waste municipal garbage mining sewage and oil spills.	CO5

4	Describe briefly the sources, effects and control of noise pollution.	Understand	Learner to recall Noise pollution mainly comes from transportation, industries, construction, domestic appliances, and public events. It can cause hearing loss, stress, sleep disorders, and health problems while also disturbing wildlife and the environment.	CO5
5	Enumerate any five major categories of water pollutants, their sources and effects.	Understand	Learner to recall five major categories of water pollutants are pathogens, organic pollutants, nutrients, toxic chemicals, and suspended solids.	CO5
6	Discuss how e-waste can be managed to prevent environmental pollution.	Understand	Learner to recall E-waste can be managed effectively by reducing, reusing, recycling, proper collection, safe disposal, producer responsibility, public awareness, and strict enforcement of environmental laws.	CO5
7	Describe the various ways of treating and purifying sewage water.	Understand	Learner to recall Treating and purifying sewage water involves preliminary primary secondary and tertiary treatment followed by sludge management.	CO5
8	What adverse effects can solid wastes cause? Discuss how can the solid waste be managed.	Understand	Learner to recall Improper disposal of solid waste causes soil, water, and air pollution, spreads diseases, harms wildlife, blocks drainage systems, and contributes to climate change.	CO5

9	Explain the concept of bioremediation. What are the advantages of bioremediation?	Understand	Learner to recall Bioremediation is a natural and eco-friendly technique that uses microorganisms or plants to clean polluted soil and water.l	CO5
10	Discuss Kyoto protocol. What can it do to curb climate change?	Understand	Learner to recall the he Kyoto Protocol is a landmark international agreement aimed at reducing greenhouse gas emissions and combating climate change.	CO5
11	What are the major causes and consequences of deforestation?	Understand	Learner to recall Deforestation is mainly caused by agriculture, urbanization, industrial activities, mining, fuel wood collection, and forest fires.	CO5
12	Discuss the difference of opinion between north block and south block countries during earth summit, 1992	Understand	Learner to recall The North-South divide at the 1992 Earth Summit was mainly about responsibility, financial support, and the right to development.	CO5
13	Explain climate change and global warming. How are they related and what is the evidence that proves climate is changing?	Understand	Learner to recall Global warming refers mainly to the rise in Earth's temperature, while climate change includes the wider effects caused by this warming.	CO5
14	What are greenhouse gases? Discuss the potential and contribution of these gases to global warming phenomenon .	Understand	Learner to recall Global warming refers mainly to the rise in Earth's temperature, while climate change includes the wider effects caused by this warming.	CO5

15	What are ozone depleting substances? Discuss the theme of Montreal protocol.	Understand	Learner to recallThe learner will be able to recall and list ozone-depleting substances and recognize their role in damaging the ozone layer.	CO5
16	Describe briefly waste water treatment methods?	Understand	Learner to The learner will be able to recall the main methods of wastewater treatment and understand their purpose in removing pollutants from wastewater before it is released back into the environment.	CO5
17	Discuss Montreal protocol. What can it do to curb climate change?.	Understand	Learner to recallThe learner will be able to recall and list ozone-depleting substances and recognize their role in damaging the ozone layer	CO5
18	Enumerate with examples the sources of surface water?	Understand	Learner to recall the The learner will be able to recall the major sources of surface water and identify different natural water bodies found on the Earth's surface.	CO5
19	What are the major causes and consequences of desertification?	Understand	Learner to recall The major causes and consequences of desertification and understand how it affects land ecosystems and human livelihoods.	CO5
20	Explain the consequences of surface and ground water?	Understand	Learner to recall The learner will be able to recall the consequences of surface water and groundwater changes and understand their effects on the environment, human health, and water availability.	CO5
PART C-SHORT ANSWER QUESTIONS				

1	Define pollution. Name various atmospheric pollutants.	Understand	Learner to recall define pollution and name common atmospheric pollutants responsible for environmental and health impacts.	CO5
2	What is noise pollution? Mention its sources.	Understand	Learner to recall meaning of noise pollution and identify its major sources that affect human health and the environment.	CO5
3	What are the natural and man-made pollutants that cause air pollution?	Understand	Learner to recall natural and man-made pollutants that cause air pollution and identify their sources.	CO5
4	Mention various methods to control air pollution in industries.	Understand	Learner to recall natural and man-made pollutants that cause air pollution and identify their sources.	CO5
5	Discuss soil pollution caused by fertilizers	Understand	Learner to recall natural and man-made pollutants that cause soil pollution and identify their sources.	CO5
6	Discuss the adverse health effects due to industrial noise.	Understand	The learner to recall and discuss the adverse health effects caused by industrial noise exposure and understand its impact on workers' health and well-being.	CO5
7	Explain how ground water get pollutes.	Understand	learner to recall how groundwater gets polluted and identify the major sources responsible for groundwater contamination.	CO5
8	Define air pollution. What are the sources of air pollution?	Understand	Learner to recall natural and man-made pollutants that cause soil pollution and identify their sources.	CO5

9	Define solid waste. How can solid waste be recycled?	Understand	Learner to recall the meaning of solid waste and explain methods of solid waste recycling to reduce environmental pollution.	CO5
10	Describe how we measure water quality	Understand	learner to recall will be how water quality is measured and identify the important parameters used to assess the suitability of water for different uses.	CO5
11	Mention treatment methods used for industrial effluents.	Understand	learner to recall the treatment methods used for industrial effluents and understand their role in removing pollutants before wastewater is discharged into the environment.	CO5
12	Define e-waste. Why does it require management	Understand	learner to recall the meaning of e-waste and explain why it requires proper management to protect human health and the environment.	CO5
13	What is bioremediation?	Understand	learner to recall the meaning of bioremediation and understand its role in controlling environmental pollution	CO5
14	What is ozone layer and why is it getting depleted?	Understand	The learner to recall the meaning of the ozone layer and explain the reasons for its depletion.	CO5
15	What are the major causes of deforestation?	Understand	Learner to recall the major causes of deforestation and understand how human activities and natural factors contribute to the loss of forest cover.	CO5

16	Outline the provisions of the Kyoto protocol.	Understand	learner to recall the major provisions of the Kyoto Protocol and understand its role in controlling greenhouse gas emissions and addressing climate change.	CO5
17	What are ozone depleting substances?	Understand	The learner to recall ozone-depleting substances (ODS) and identify the chemicals responsible for damaging the ozone layer.	CO5
18	What are the major causes of desertification?	Understand	learner to recall the major causes of desertification and understand the factors that lead to land degradation and loss of soil productivity.	CO5
19	Differentiate between climate change and global warming.	Understand	learner to recall the difference between climate change and global warming and understand their relationship.	CO5
20	What are the results of implementation of the Montreal protocol	Understand	learner to recall the implementation of the Montreal Protocol and understand its role in protecting the ozone layer.	CO5
MODULE V				
ENVIRONMENTAL POLICY AND LEGISLATION				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1	Why do we refer environment protection act 1986 as an umbrella act? Discuss the role of various governmental agencies in environmental protection and control.	Understand	Learner to recall why the Environment Protection Act, 1986 is referred to as an “umbrella act” and understand its role in environmental protection.	CO6

2	How do you define the pollution as per water act, 1974? Write the salient features of this act.	Understand	Learner to recall the definition of pollution as per the Water Act, 1974, and identify the salient features of the Act.	CO6
3	Explain the concept of green building. How will green building impact the environment overall.	Understand	Learner to recall The learner will be able to recall the concept of green buildings and explain their overall impact on the environment..	CO6
4	What are the rules and regulations that guide waste management in India?	Understand	Learner to recall The learner will be able to recall the major rules and regulations that guide waste management in India and understand their role in proper waste handling and environmental protection.	CO6
5	Why is sustainable development so often associated with protecting the environment?.	Understand	Learner to recall the concept of sustainable development and explain why it is often associated with protecting the environment.	CO6
6	Explain how does wild life protection act, 1972 deal with human-animal conflict? Mention the major threats being faced by the wild life in India.	Understand	Learner to recall how the Wildlife Protection Act, 1972 addresses human-animal conflict and understand its role in conserving wildlife while reducing conflicts between humans and animals.	CO6
7	What is meant by a basic environmental assessment? Describe the key steps of this process.	Understand	Learner to recall the concept of basic environmental assessment and describe the key steps involved in the process.	CO6

8	Why are developing countries experiencing rapid population growth while developed countries are growing more slowly?	Understand	Learner to recall the differences in population growth patterns between developing and developed countries and understand the factors influencing these trends.	CO6
9	What are the functions of central board and state board under the air act?	Understand	Learner to recall the characteristics of lubricants and understand the properties of a good lubricant.	CO6
10	Environmental education can play an important role in environmental protection". Explain it.	Understand	Learner to recall the importance of environmental education and understand its role in environmental protection.	CO6
PART B-LONG ANSWER QUESTIONS				
1	Discuss the salient features of Air (Prevention and Control of Pollution) Act, 1981.	Understand	Learner to recall the key provisions of the Air Act, 1981 and explain its importance in maintaining air quality and protecting public health.	CO6
2	What are the strategies for making cities and communities sustainable	Understand	Learner to recall key strategies for sustainable urban development and explain how they help protect the environment and improve community well-being.	CO6
3	Discuss the salient features of Environmental protection Act, 1986.	Understand	Learner to recall salient features of the Environment (Protection) Act, 1986 and understand its role in protecting and improving the environment.	CO6

4	List major provisions in Forest Conservation Act, 1980.	Understand	Learner to recall the major provisions of the Forest (Conservation) Act, 1980 and understand its role in protecting forest resources in India.	CO6
5	Discuss the salient features of Wild life protection Act..	Understand	Learner to recall the salient features of the Wildlife Protection Act, 1972 and understand its role in conserving wildlife and their habitats in India.	CO6
6	What are the major municipal solid waste management and handling rules?	Understand	Learner to recall the key provisions of Municipal Solid Waste Management Rules and understand their importance in reducing waste pollution and promoting sustainable waste management.	CO6
7	Define biomedical wastes. What are the rules to manage and handle them?	Understand	Learner to recall concept of biomedical waste and understand its sources and importance of proper management.	CO6
8	What are hazardous wastes? Discuss the rules to manage and handle them.	Understand	Learner to recall concept of hazardous waste and discuss the rules for its management and handling to prevent risks to human health and the environment.	CO6
9	Define sustainable development. What are the threats and measures for sustainable development.	Understand	Learner to recall the concept of sustainable development and identify the major threats and measures for achieving sustainable development.	CO6

10	Describe the importance of environmental education. What is value-based environmental education?	Understand	Learner to recall the importance of environmental education and explain the concept of value-based environmental education.	CO6
11	Write the objectives, principles, key elements and importance of environmental impact assessment.	Understand	Learner to recall the objectives, principles key elements and importance of Environmental Impact Assessment (EIA) and understand its role in sustainable development.	CO6
12	Discuss the major components of population growth. What are the advantages of having a healthy population?	Understand	Learner to recall the major components of population growth and explain the advantages of having a healthy population.	CO6
13	Explain how can extreme consumerism among a society's wealthiest members affect middle-class people?	Understand	Learner to recall how extreme consumerism among a society's wealthiest members can affect middle class people and understand its social, economic, and environmental impacts.	CO6
14	What is urban sprawl? Mention causes and effects of urban sprawl.	Understand	Learner to recall the concept of urban sprawl and identify its major causes and effects.	CO6
15	Explain the concept of green building. How will green building impact the environment overall.	Understand	Learner to recall the concept of green buildings and explain their overall impact on the environment.	CO6
16	Discuss the salient features of Water Act.	Understand	Learner to recall the salient features of the Water (Prevention and Control of Pollution) Act, 1974 and understand its role in preventing and controlling water pollution.	CO6

17	Write the objectives, principles, key elements and importance of environmental education..	Understand	Learner to recall the objectives principles key elements and importance of environmental education and understand its role in environmental protection and sustainable development.	CO6
18	Discuss the role of government towards the sustainable development?	Understand	Learner to recall the role of the government in achieving sustainable development and understand how government policies and actions contribute to environmental social and economic well-being.	CO6
19	What is population explosion? Write the consequences and controlling methods of population explosion.	Understand	Learner to recall the concept of population explosion and explain its consequences and methods of control.	CO6
20	Explain briefly the concept of crazy consumption and its effects ?.	Understand	Learner to recall the concept of crazy consumption and explain its effects on society and the environment.	CO6
PART C-SHORT ANSWER QUESTIONS				
1	List out the objectives of Air pollution act.	Understand	Learner to recall the objectives of the Air (Prevention and Control of Pollution) Act, 1981 and understand its role in controlling air pollution.	CO6
2	What are the objectives of wild life protection act?	Understand	Learner to recall the main objectives of the Wildlife Protection Act, 1972 and explain its importance in conserving wildlife and maintaining ecological balance.	CO6

3	Enlist various acts related to environment protection.	Understand	Learner to recall and identify the major environmental protection laws and their purposes.	CO6
4	What is the role of green buildings in reducing global warming?	Understand	Learner to recall Green buildings are designed constructed and operated to reduce environmental impact.	CO6
5	Write a note on bio-medical wastes.	Understand	Learner to recall Biomedical waste is waste generated during the diagnosis, treatment, or immunization of humans or animals, as well as during medical research and healthcare activities.	CO6
6	What is meant by environmental impact assessment	Understand	Learner to recall Environmental Impact Assessment (EIA) is a systematic process used to identify predict and evaluate the potential environmental effects of a proposed project before it is approved.	CO6
7	What is meant by crazy consumerism?.	Understand	Learner to recall Crazy consumerism refers to excessive and unnecessary buying of goods and services beyond actual needs.	CO6
8	Enumerate the various effects of urban sprawl.	Understand	Learner to recall Urban sprawl is the uncontrolled expansion of cities into surrounding rural or natural areas.	CO6
9	What is the importance of environmental education?.	Understand	Learner to recall Environmental education helps people understand the importance of protecting the environment and using natural resources responsibly.	CO6

10	List any three categories of waste and give their sources..	Understand	Learner to recall categories of waste and give their sources.	CO6
11	What does the forest conservation act of 1980 specify?	Understand	Learner to recall The Forest (Conservation) Act, 1980 aims to conserve forests and regulate the use of forest land.	CO6
12	Define hazardous waste. Are hazardous wastes dumped in India?	Understand	Learner to recall Hazardous waste is waste that is toxic, corrosive, flammable, reactive, or otherwise harmful to human health and the environment.	CO6
13	Mention some salient features of biomedical waste management and handling rules, 2016.	Understand	Learner to recall	CO6
14	What is municipal solid waste and why does it require management?.	Understand	Learner to recall Municipal Solid Waste (MSW) refers to the solid waste generated from households, shops, offices, schools, institutions, markets, and other community activities.	CO6
15	Define sustainable development. Why is it necessary?	Understand	Learner to recall Sustainable development is the process of meeting the needs of the present generation without compromising the ability of future generations to meet their own needs	CO6
16	What is meant by population explosion?	Understand	Learner to recall Population explosion refers to the rapid and excessive increase in the number of people in a particular area or on Earth within a short period of time.	CO6

17	Discuss the basic characteristics of green buildings.	Understand	Learner to recall green buildings are structures designed to reduce environmental impact and use resources efficiently.	CO6
18	What are some threats to sustainability?	Understand	Learner to recall Threats to sustainability are activities and conditions that harm the environment, reduce natural resources, and affect the ability of future generations to meet their needs.	CO6
19	What are the duties and powers assigned to State Pollution Control Board under water act?	Understand	Learner to recall The State Pollution Control Board (SPCB) is responsible for preventing and controlling water pollution and maintaining the quality of water resources.	CO6
20	Mention the objectives of environmental protection act.	Understand	Learner to recall The Environment (Protection) Act, 1986 was enacted to protect and improve the environment and to prevent hazards to humans, other living organisms, and property.	CO6

Course Coordinator:
Dr.M.S.N.A.Prasad, Associate Professor

HOD ,CSE